

Supplement Table 1 Framework of PubChemFP Model with best performance

input_1 (InputLayer)	[(None, 27, 27, 1)]	0
flatten (Flatten)	(None, 729)	0
dense (Dense)	(None, 1024)	747520
dense_1 (Dense)	(None, 512)	524800
dense_2 (Dense)	(None, 128)	65664
dense_3 (Dense)	(None, 64)	8256
dense_4 (Dense)	(None, 32)	2080
dense_5 (Dense)	(None, 3)	99
dense_6 (Dense)	(None, 32)	128
dense_7 (Dense)	(None, 64)	2112
dense_8 (Dense)	(None, 128)	8320
dense_9 (Dense)	(None, 512)	66048
dense_10 (Dense)	(None, 1024)	525312
dense_11 (Dense)	(None, 729)	747225
reshape (Reshape)	(None, 27, 27, 1)	0
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Total params: 2,697,564		
Trainable params: 2,697,564		
Non-trainable params: 0		

The first column represents input data, the numbering of fully connected layers, and output data. The second column displays the dimensions of input data, the node count of fully connected layers, and the dimensions of output data, with node counts such as "1024," "512," "128," etc. The last column indicates the parameter count required for connections between preceding and subsequent fully connected layers. Each Dense layer is followed by a activation 'ReLU' layer until the last reshape layer is preceded by a 'Sigmoid' activation layer.

Supplement Table 2 Framework of MACCSFP Model with best performance

flatten (Flatten)	(None, 169)	0
dense (Dense)	(None, 512)	87040
dense_1 (Dense)	(None, 1024)	525312
dense_2 (Dense)	(None, 512)	524800
dense_3 (Dense)	(None, 256)	131328
dense_4 (Dense)	(None, 64)	16448
dense_5 (Dense)	(None, 32)	2080
dense_6 (Dense)	(None, 3)	99
dense_7 (Dense)	(None, 32)	128
dense_8 (Dense)	(None, 64)	2112
dense_9 (Dense)	(None, 256)	16640
dense_10 (Dense)	(None, 512)	131584
dense_11 (Dense)	(None, 1024)	525312
dense_12 (Dense)	(None, 512)	524800
dense_13 (Dense)	(None, 169)	86697
reshape (Reshape)	(None, 13, 13, 1)	0
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Total params: 2,574,380		

Supplement Table 3 Framework of PharmacoPFP Model with best performance

flatten (Flatten)	(None, 289)	0
dense (Dense)	(None, 512)	148480
dense_1 (Dense)	(None, 1024)	525312
dense_2 (Dense)	(None, 512)	524800
dense_3 (Dense)	(None, 256)	131328
dense_4 (Dense)	(None, 64)	16448
dense_5 (Dense)	(None, 32)	2080
dense_6 (Dense)	(None, 3)	99
dense_7 (Dense)	(None, 32)	128
dense_8 (Dense)	(None, 64)	2112
dense_9 (Dense)	(None, 256)	16640
dense_10 (Dense)	(None, 512)	131584
dense_11 (Dense)	(None, 1024)	525312
dense_12 (Dense)	(None, 512)	524800
dense_13 (Dense)	(None, 289)	148257
reshape (Reshape)	(None, 17, 17, 1)	0
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Total params: 2,697,380		

Supplement Table 4 Comparison of PubChemFP Parameters and Results

Parameter Number	Encoder Network Depth	Encoder Structure	Total Parameter Count	MSE
Parameter 1 (best)	5	1024-512-128-64-32	2697564	0.0245
Parameter 2	6	2048-1024-512-128-64-32	8388956	0.1400
Parameter 3	8	2048-1024-512-128-64-32-16-8	8390140	0.0750
Parameter 4	9	2048-1024-512-256-128-64-32-16-8	8587260	0.0690
Parameter 5	8	1024-512-256-128-64-32-16-8	2895868	0.0400
Parameter 6	7	1024-512-256-128-64-16-8	2892732	0.0525
Parameter 7	7	2048-1024-256-128-64-32-8	7799004	0.0800
Parameter 8	7	2048-1024-512-128-64-32-8	8389340	0.0848
Parameter 9	4	512-128-64-32	900444	0.0269

Supplement Table 5 Comparison of MACCSFP Parameters and Results

Parameter Number	Encoder Network Depth	Encoder Structure	Total Parameter Count	MSE
Parameter 1	5	1024-512-128-64-32	1550124	0.0555
Parameter 2	5	256-128-64-32-16	174828	0.0520
Parameter 3	5	512-512-256-64-16	997292	0.0455
Parameter 4	6	256-512-256-128-64-16	697516	0.0475
Parameter 5	6	512-1024-512-128-64-16	2424492	0.0455
Parameter 6	6	512-1024-512-256-64-16	2572204	0.0430
Parameter 7	6	512-1024-512-128-64-32	2426668	0.0462
Parameter 8	6	512-1024-512-128-64-32	2426668	0.0440
Parameter 9 (best)	6	512-1024-512-256-64-32	2574380	0.0390

Supplement Table 6 Comparison of PharmacoPFP Parameters and Results

Parameter Number	Encoder Network Depth	Encoder Structure	Total Parameter Count	MSE
Parameter 1 (best)	6	512-1024-512-256-64-32	2697380	0.0245
Parameter 2	6	1024-512-512-256-64-32	2469028	0.0250
Parameter 3	5	1024-512-256-64-32	1943716	0.0260
Parameter 4	5	512-512-256-64-32	1122468	0.0260